

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:4

AIM: 4. Perform basic Image Handling and processing operations on the image

- **Read an image in python and Dilate an Image using Dilate function.**

Algorithm:

- Import libraries and read image: import cv2, numpy; load with cv2.imread(path, flag) and check for None.
- Convert or prepare mask: if needed, convert to grayscale (cv2.cvtColor) and optionally threshold to obtain a binary mask.
- Choose structuring element: pick shape (cv2.MORPH_RECT / MORPH_ELLIPSE / MORPH_CROSS) and size (e.g., (3,3), (5,5)).
- Create kernel: kernel = cv2.getStructuringElement(shape, (kx, ky)).
- Apply dilation: result = cv2.dilate(src, kernel, iterations=n) where iterations controls expansion strength.
- Save or display and cleanup: cv2.imwrite(...) or cv2.imshow(...); then cv2.destroyAllWindows().

Code:

```
import cv2
import numpy as np

img = cv2.imread(r"C:\Users\surya\OneDrive\Pictures\Screenshots\1\Screenshot 2026-07-23 113933.png")
if img is None:
    print("Error: Image not found or unable to open.")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.ones((3, 3), np.uint8)
```

```
dilated = cv2.dilate(gray, kernel, iterations=1)
```

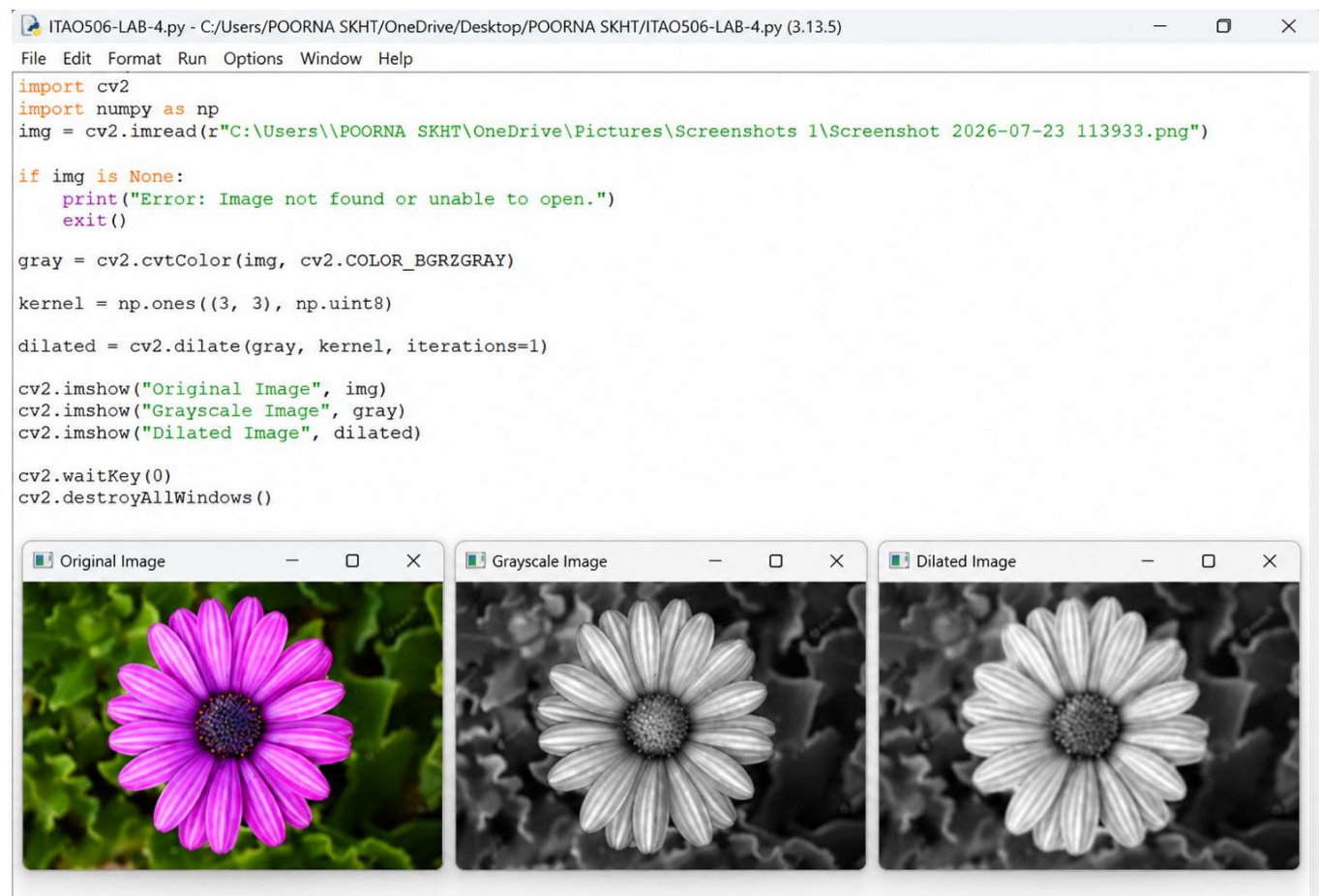
```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Grayscale Image", gray)
```

```
cv2.imshow("Dilated Image", dilated)
```

```
cv2.waitKey(0) cv2.destroyAllWindows()
```

OUTPUT :



Result

Thus, the input image was successfully read and processed using the **Dilate** function in Python with OpenCV. The dilation operation expanded the boundaries of the objects in the image, making the edges thicker and enhancing the prominent features successfully.